MILITARY SPECIFICATION

ANTENNA, FIXED, HIGH FREQUENCY, VERTICAL POLARIZATION, OMNIDIRECTIONAL, BROADBAND

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the antenna described herein shall consist of this specification and the latest issue of MIL-A-28768.

1. SCOPE

1.1 Scope. This specification covers requirements for a fixed, high-frequency (HF), vertical, polarization, broadband, omnidirectional antenna, hereinafter referred to as the antenna.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standard(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto cited in the solicitation.

SPECIFICATIONS

FEDERAL

W-R-550	Rods, Ground (With Attachments)
QQ-C-576	Copper Flat Products With Slit, Slit and Edge-rolled, Sheared, Sawed, Or Machined Edges (Plate, Bar, Sheet, And Strip)

MILITARY

MIL-A-28768	Antennas, Fixed, High Frequency, General Specification For

STANDARD

MILITARY

MIL-STD-1364	Standard General Purpose Electronic Test Equipment

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations, unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-003-121), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5985

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3. REQUIREMENTS

3.1 General. The antenna shall comply with the requirements of MIL-A-28768 and the requirements specified herein. The antenna shall be furnished as a complete system ready for installation and operation. The antenna shall be designed for broadband operation without tuning or adjustment for operating frequency.

3.2 Size. The antenna shall comply with the size limitations specified in TABLE I.

TABLE I. Size of antenna.

Maximum height of antenna structure	95 ft 1/
Maximum included installation area of antenna structure (including guys and anchors, but not including ground system)	32,670 ft^2 2/
Maximum length, width, diagonal, or diameter of included installation area of antenna structure (including guys and anchors, but not including ground system)	204 ft
Antenna ground system radius	123 ft

1/ Feet
2/ Square feet

3.3 Classification. The antenna shall be classified as specified in 3.3.1 and 3.3.2.

3.3.1 Power input capability. The antenna shall be of the power input capability Power 10 as specified in the Power input capability paragraph of MIL-A-28768.

3.3.2 Wind class. The antenna shall be of the Wind Class 2 as specified in the Wind classes paragraph of MIL-A-28768.

3.4 Environmental conditions. The antenna shall conform to the environmental requirements specified in MIL-A-28768. The antenna shall be designed so that snow and ice will not accumulate at the antenna feedpoint and at the base of the antenna to the extent that the allowable antenna stress requirements specified in the Structural requirements paragraph of MIL-A-28768 are exceeded or that the antenna electrical requirements are degraded, or both. Structural relief devices (fuses) shall not be used in the antenna to comply with the environmental conditions requirements.

3.5 Structural requirements. The antenna shall conform to the structural requirements specified in 3.5.1 through 3.5.5.

3.5.1 Rotational stability. The antenna shall be designed with an appropriate guying or bracing system which shall eliminate the possibility of rotational instability. The antenna shall be designed so that rotation about a vertical axis through the center of the antenna shall be not greater than 30 minutes of arc under the environmental conditions specified in 3.4.

3.5.2 Field adjustment of cable tension. The design of the antenna shall provide the capability for field adjustment of cable tensions both during and after erection.

3.5.3 Antenna installation design. The antenna shall be designed so that installation grading will not be required within the area of the antenna ground system if the ground roughness is within ±3 ft and the overall ground slope is within ±3 degrees. Antenna support tower(s) shall remain vertical after installation of the antenna.

3.5.4 Installation materials. All additional parts and materials (such as guy material, anchors, and attachment hardware) which are necessary to permit antenna installation over a site ranging from flat and level to the limits specified in 3.5.3 shall be considered a part of and supplied with the antenna.

3.5.5 Wooden poles. Wooden poles shall not be used or required.

3.6 Electrical performance. The antenna shall comply with the design requirements of the Electrical design paragraph of MIL-A-28768, except as otherwise specified in 3.6.1 through 3.6.6.

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3.6.1 Frequency range. The antenna shall operate over the frequency range of from 2.0 megahertz (MHz) to 30 MHz.

3.6.2 Input impedance. The antenna shall conform to the requirements of the Input impedance paragraph of MIL-A-28768, except that the voltage standing wave ratio (VSWR) shall not exceed 6:1 with respect to 50 ohms from 2.0 MHz to 3.999999 MHz and shall not exceed 2.5:1 with respect to 50 ohms from 4.0 MHz to 30 MHz.

3.6.3 Polarization. The antenna shall radiate electromagnetic waves that are vertically polarized.

3.6.4 Azimuth radiation. With the antenna installed over perfectly conducting earth, the radiation pattern of the antenna at each frequency in the frequency range (see 3.6.1) from 0 degrees azimuth to 360 degrees azimuth at, in turn, each elevation angle from 0 degrees to 20 degrees shall be omnidirectional to within ±1 decibel (dB).

3.6.5 Directivity. With the antenna installed over perfectly conducting earth, the antenna directivity at the elevation plane antenna radiation maximum shall be not less than +4.8 dB with respect to an isotropic radiator in free space (dBi) at each frequency in the frequency range (see 3.6.1). With the antenna installed over perfectly conducting earth, the antenna directivity at elevation angles in the range from 0 degrees (at the horizon) to +10 degrees (above the horizon) shall be not less than +2 dBi at each frequency in the frequency range (see 3.6.1).

3.6.6 Antenna efficiency. When installed over imperfectly conducting earth, the efficiency of the antenna, as specified in the Definitions paragraph of MIL-A-28768, shall be not less than 75 percent from 2.0 MHz to 3.999999 MHz and shall be not less than 85 percent from 4.0 MHz to 30 MHz.

3.7 Ground system. An antenna ground system shall be supplied as specified in MIL-A-28768 and as follows. The ground system shall consist of radials extending from the center (vertical) axis of the antenna. A minimum of 120 ground system radials per antenna shall be required. The wires shall be equally spaced in azimuth at 3-degree (±0.5 degree) intervals. The length of each radial shall be as specified in TABLE I. At the outer extremity of the ground system, every third radial shall be terminated to a ground rod, conforming to Type II, Class B of W-R-550, except that the diameter shall be 0.625 inches (in.) (+0, -0.0781 in.) and the length shall be 6 ft (±0.750 in.). A circumferential wire of the same gage and characteristics as the radials shall be supplied for bonding to the ends of the unterminated radials and to the ground rods. The inner termination of each radial shall be to a copper plate conforming to QQ-C-576. The plate may be either circular or rectangular, shall be minimum of 7 ft^2 in area by 0.0647 in. thick, and shall cover the top and overlap the sides of the base support for the antenna. The radials shall maintain separation up to their point of attachment to the plate. The inner and outer attachments of the radial wires shall be designed to be made in the field utilizing brazing. The inner attachment shall be above ground level and shall be holes in, or eyes on, the plate to attach the wires so that the braze is not the sole means of bonding the wires to the plate. The radials need not be precut or preassembled. All materials necessary to make the connections shall be supplied. For distances less than 25 ft from the center of the antenna the radials shall be designed to be placed on the earth's surface. For distances greater than 25 ft from the center of the antenna the radials shall be designed to be buried to a maximum of 6 in.

3.8 General purpose electronic test equipment (GPETE). The antenna shall be designed so that all testing can be accomplished using GPETE selected from MIL-STD-1364. Any nonstandard and special purpose electronic test equipment, if necessary for antenna testing, shall be subject to the approval of the procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions. Quality assurance provisions shall be in accordance with MIL-A-28768 and as specified in 4.2 and 4.3.

4.2 First article inspection. Unless otherwise specified herein (see 6.2), one antenna shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the MIL-A-28768 examinations and tests specified in TABLE II and in a through g:

a. Measurement of installed antenna dimensions to verify compliance with the size requirements of 3.2
b. Analyses and calculations to demonstrate compliance with the rotational stability requirements of 3.5.1

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c. Analyses and calculations to demonstrate compliance with the installation grading requirements of 3.5.3
d. Measurement of the antenna input VSWR in accordance with the Impedance and VSWR paragraph of MIL-A-28768 to verify compliance with the frequency range and input impedance requirements of 3.6.1 and 3.6.2, respectively
e. Analyses and calculations to demonstrate compliance with the frequency range and azimuth radiation pattern requirements of 3.6.1 and 3.6.4, respectively
f. Analyses and calculations to demonstrate compliance with the antenna directivity requirements of 3.6.5
g. Analyses and calculations to demonstrate compliance with the antenna efficiency requirements of 3.6.6

4.3 Production inspection. Production inspection shall be conducted on each antenna offered for delivery. This inspection shall comprise such examinations and tests as will prove the workmanship and reveal omissions and errors in the production process. Production inspection shall include the MIL-A-28768 examinations and tests specified in TABLE II.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preparation of delivery shall be as specified in MIL-A-28768.

6. NOTES

6.1 Intended use. The antenna will be installed in U.S. Navy shore transmitting sites for operation with a 5 watt to 200 watt swept frequency (2.0 MHz to 30 MHz) frequency modulation and continuous wave transmitter (that is, the T-1373/TRQ-35(V) transmitter) used to sound HF radio propagation paths out to distances of 2000 miles from the antenna. However, the antenna must also be able to operate over the 4.0 MHz to 30 MHz frequency range with a 10 kilowatt HF transmitter which can also be installed at U.S. Navy shore transmitting sites.

6.2 Ordering data. In addition to the items specified in MIL-A-28768, procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than as specified in 4.2.

6.3 First article. When a first article inspection is required, the item(s) should be a first production sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article tests results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

TABLE II. MIL-A-28768 examinations and tests.

Examination or test	MIL-A-28768 test paragraph	First article inspection	Production inspection
Surface examination:			
Parts and materials	4.5.1	X	X
Finish	4.5.1	X	X
Marking	4.5.1	X	X
Safety	4.5.11	X	X
Workmanship	4.5.1	X	X
Weld	4.5.7	X	X
Paint system	4.5.1	X	X
Inclined plane	4.5.2	X	

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TABLE II. MIL-A-28768 examinations and tests. (continued)

Examination or test	MIL-A-28768 test paragraph	First article inspection	Production inspection
Special tests of nonmetallic guys and assemblies:			
Breaking strength in tension	4.5.3.1	X	
Wet strength	4.5.3.2	X	
Fatigue strength	4.5.3.3	X	
Elongation	4.5.3.4	X	
Proof loading:			
Cable elements	4.5.4.1	X	
Tower climbing device	4.5.4.2	X	
Lifting attachments	4.5.4.3	X	
Enclosure	4.5.5	X	
Installability	4.5.6	X	
Electrical design tests:			
Direct current continuity to ground and contact resistances	4.5.8.2	X	
RF power	4.5.8.4	X	
Thermal design	4.5.9	X	
Environmental tests:			
Fungus resistance	4.5.10.1	X	
Humidity	4.5.10.2	X	
Salt fog	4.5.10.3	X	
Electromagnetic pulse	4.5.10.5	X	
Maintenance demonstration	4.6	X	
Inspection of preparation for delivery	4.7	X	

Preparing activity:
NAVY - EC

(Project No. 5985-N551(EC))

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